

Coding & Solution

Date	08 October 2026
Team ID	UNM1611GUFRAN/SWTID-2026-9802
Project Name	AI FAQ Assistant API
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/nethrasoundarrajan02-star/API-FAQ
Programming Language(s)	Java script (Node.js)
Framework(s) Used	Express.js, Mongoose
Key Features Implemented	1. User Registration and Login with JWT Authentication and bcrypt password hashing 2. User Profile Management 3. FAQ CRUD Operations 4. FAQ Category Validation 5. Keyword-based FAQ Search using MongoDB Regex 6. AI Answer Generation using Google Gemini (gemini-2.0-flash) 7. AI-powered Structured FAQ Generation 8. Protected API Endpoints with JWT Middleware 9. Input Validation and Centralized Error Handling 10. MVC Architecture
Pending / Incomplete Features	Admin dashboard, multi-language support, advanced FAQ analytics, and additional AI-powered features are planned for future development.
Setup / Run Instructions	1. Clone the repository. 2. Run npm install. 3. Create a .env file with PORT, MONGO_URI, JWT_SECRET, and GEMINI_API_KEY. 4. Start MongoDB. 5. Run npm run dev for development or npm start for production. 6. Test the API using Postman with the provided collection.

Code Quality Checklist

S.N o	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The AI FAQ Assistant API is implemented in Python using FastAPI with a modular structure (separate routes, services and models). The core features run without critical errors and the source code is committed to a version control repository. Remaining work includes multi-language support and an admin dashboard for managing FAQs.